Introduction: from belief sharing to robust generalized Bayes

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Introduction: from belief sharing to robust generalized Bayes

A colony of active-inference agents that shares beliefs inherits both the power and the fragility of the pool it uses: the same multiplication of reports that sharpens an honest consensus lets a single confident, wrong member capture it.

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Introduction (part 2)

Two research communities each hold part of the remedy but do not, on their own, close the gap.

Introduction (part 3)

The active-inference community gives agents generative models, action selection, and colony-level belief sharing. The robust- and federated-Bayes community gives generalized objectives and aggregation methods for inference under misspecification or contamination.

Introduction (part 4)

The cited literatures do not, however, provide a single tested treatment of robust categorical belief fusion for active-inference agents. This introduction states the problem and its evidence boundary; sec. 6 scopes the reviewed gap,

Introduction (part 5)

and sec. 7 lists the contributions together with what each result does not establish.

Introduction (part 6)

The opening graphical abstracts provide a reading map: the generative model and belief-exchange cycle in fig. 1, the distinct client and server robustness boundaries in fig. 2,

Introduction (part 7)

and the model-to-execution evidence chain in fig. 3. These conceptual schematics orient the detailed assumptions, methods, and evidence that follow.

Active inference supplies generative agents and shared beliefs

Active inference casts perception, learning, and action as the minimization of variational free energy under a generative model,

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Active inference supplies... (part 2)

a program that grew out of the free-energy principle (Friston 2010) and its process-theory implementation (Friston et al. 2017).

Active inference supplies... (part 3)

In the discrete state-space setting the formalism is now standardized. An agent carries the categorical tensors A (likelihood), B (transitions), C (preferences), and D_0 (initial priors), and infers hidden states by minimizing variational free energy. It selects actions by minimizing *expected* free energy —

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Active inference supplies... (part 4)

the sum of a risk term and an ambiguity term that together trade off goal-seeking against uncertainty-resolving behavior (Da Costa et al. 2020).

Active inference supplies... (part 5)

This synthesis is the substrate we build on. Mature toolboxes make it executable at scale: pymdp provides discrete-state active inference in Python (Heins et al. 2022), and RxInfer delivers reactive message passing for exact Bayesian inference (Bagaev et al. 2023).

Active inference supplies... (part 6)

The community has also moved from single agents to *collectives*. Surprise-minimizing ensembles reproduce collective animal behavior such as schooling and flocking (Heins et al. 2024); epistemic communities form when agents share a generative model and exchange evidence (Albarracin et al. 2022);

Active inference supplies... (part 7)

and collective intelligence has been framed directly in active-inference terms (Kaufmann et al. 2021).

Active inference supplies... (part 8)

The narrow bridge studied here starts from posterior broadcasts over one shared categorical factor — a predator's location, say — and forms a project log-linear pool (eq. 9). Under the explicit finite-shared-support, posterior-log-potential,

Active inference supplies... (part 9)

and fixed-weight assumptions of sec. 8.8, that weighted geometric pool is a specialization of Friston et al.'s Eq. 7 message-combination term (Friston et al. 2024).

Active inference supplies... (part 10)

It is not a reconstruction of the source message construction, scheduling, cavity policy, generative factors, or complete protocol. That representation connects the qualified categorical bridge to the classical logarithmic-pooling literature (Genest and Zidek 1986; Genest et al. 1986),

Active inference supplies... (part 11)

to modern Bayesian treatments of log-pool weights (Carvalho et al. 2023), to product-of-experts geometry in machine learning (Hinton 2002), and to distributed Bayesian estimators (Tresp 2000). In the reproduced baseline,

Active inference supplies... (part 12)

communication lowers mean free energy relative to the matched incommunicado condition (sec. 10.2).

Active inference supplies... (part 13)

Friston et al. (2024) crystallized the colony mechanism into three worked simulations — communicating-colony free-energy convergence, Dirichlet language acquisition, and Bayesian model reduction structure emergence —

Active inference supplies... (part 14)

whose mechanisms motivate three reduced categorical analogues in this repository (sec. 10.1). They are not numerical or exact protocol replications; a source-parity reconstruction remains future work before evaluating source-level equivalence.

Active inference supplies... (part 15)

Alongside fusion, the community has tools for growing the model itself: active inference connects naturally to Bayesian optimal experimental design and model selection (Smith et al. 2022),

Active inference supplies... (part 16)

and post-hoc Bayesian model reduction prunes redundant structure by comparing free-energy bounds (Friston and Penny 2011) — the algebra used by our configured BMR sign control (sec. 10.4).

Active inference supplies... (part 17)

The reviewed active-inference belief-sharing work does not systematically characterize fusion under explicit contamination or intentionally wrong belief broadcasts.

Active inference supplies... (part 18)

The log-linear pool assumes reports are compatible with the shared generative model, while the opinion-pooling literature makes the assumptions about independence, weights, and external Bayesian coherence explicit (Genest and Zidek 1986; Genest et al. 1986). Fully Bayesian aggregation sharpens the point:

Active inference supplies... (part 19)

geometric pooling is normatively compelling under dynamic-Bayesian rationality assumptions, not an assumption-free robustness procedure (Dietrich 2021).

Active inference supplies... (part 20)

The same product geometry can be brittle: if a report assigns zero or near-zero mass to the true state, the product can assign near-zero mass there too. That is the failure mode tested here, not a claim that every federation or every product pool behaves identically.

Robust and federated Bayes supplies bounded-influence updating

A separate literature studies bounded-influence updating outside active inference. **Federated learning** aggregates models trained on decentralized data without pooling the data itself,

Robust and federated Bayes... (part 2)

the canonical algorithm being FedAvg (McMahan et al. 2017).

Robust and federated Bayes... (part 3)

Cast probabilistically, federated and continual learning unify under **partitioned variational inference**, in which each client owns a factor of a global approximate posterior and the server combines factors in natural-parameter space (Bui et al. 2018; Ashman et al. 2022).

Robust and federated Bayes... (part 4)

This is a closely related factor algebra expressed in a different vocabulary; the implementation tests the correspondence in the categorical recovery limit rather than assuming that the two settings are interchangeable.

Robust and federated Bayes... (part 5)

The robustness this paper needs comes from **generalized Bayesian inference**, which replaces the likelihood with a *loss* and the KL regularizer with a *general divergence*,

Robust and federated Bayes... (part 6)

so that the posterior minimizes a generalized objective rather than applying Bayes' rule literally (Bissiri et al. 2016; Jewson et al. 2018; Knoblauch et al. 2022).

Robust and federated Bayes... (part 7)

This includes Gibbs-posterior updates (Jiang and Tanner 2008), coarsened or tempered posteriors that condition on neighborhoods rather than exact data (Miller and Dunson 2018),

Robust and federated Bayes... (part 8)

and learning-rate/temperature choices designed for safe updating under misspecification (Grünwald 2012; Kleijn and Vaart 2012). Choosing a loss or divergence with a bounded-influence property — for example,

Robust and federated Bayes... (part 9)

the density-power β-loss (Basu et al. 1998; Fujisawa and Eguchi 2008; Ghosh and Basu 2015) or generalized cross-entropy (Zhang and Sabuncu 2018) — can cap the influence of a contaminated observation under the corresponding assumptions.

Robust and federated Bayes... (part 10)

FedGVI (Mildner et al. 2025) federates this idea: each client runs a robust generalized-Bayes update against a *cavity* (the global posterior with the client's own factor removed), and the server aggregates the refreshed factors under a chosen server divergence.

Robust and federated Bayes... (part 11)

The result is federated inference with divergence- and loss-specific robustness guarantees, demonstrated on Bayesian neural networks under label contamination.

Robust and federated Bayes... (part 12)

Crucially, standard Bayes is a *corner* of this generalized family — the case where the loss is the negative log-likelihood and the divergence is KL. Friston et al. do not claim FedGVI, generalized Bayes, β-divergence, or robustness; this manuscript supplies the recasting.

Robust and federated Bayes... (part 13)

Robust generalized Bayes therefore does not replace exact-Bayes belief fusion; it contains the standard pool as a tested zero-robustness recovery limit. That containment is the hinge this paper tests,

Robust and federated Bayes... (part 14)

and the recovery limits are stated formally as numbered results in sec. 9 and the central identity in sec. 8.8.

Robust and federated Bayes... (part 15)

The historical framing is deliberately modest. The manuscript uses early probability, inverse-probability, utility, and collective-judgment sources as a conceptual genealogy for belief, evidence, expectation,

Robust and federated Bayes... (part 16)

and aggregation (Pascal and Fermat 1654; Huygens 1657; Bernoulli 1713; Moivre 1718; Bayes 1763; Laplace 1774; Bernoulli 1738; Condorcet 1785). These sources do not anticipate KL divergence, product-of-experts learning, variational Bayes, or federated optimization;

Robust and federated Bayes... (part 17)

the modern correspondence to FedGVI is the formal construction proved and tested here.

Robust and federated Bayes... (part 18)

Beyond this core, we evaluate a tempered objective family, a deterministic MLP aggregation transfer, and disjoint-field-of-view communication. These are boundary tests: the first probes the accuracy/weight-control trade-off, the second tests API portability in one additional model class,

Robust and federated Bayes... (part 19)

and the third separates a binary-complement null result from a larger-state-space case where communication materially improves consensus.

Questions, design, and evidence boundary

The paper answers four scoped questions. First, does turning robustness off recover the standard log-linear pool and closed-form Bayes update? Second, under the declared confident-wrong broadcast mechanism,

Questions, design, and... (part 2)

how does the server heuristic change consensus accuracy across contamination rates? Third, what does the objective-backed variational server rule guarantee, and what accuracy does it trade away? Fourth,

Questions, design, and... (part 3)

how do communication, hierarchy, sensitivity, and parameter recovery behave in the accompanying categorical extensions?

Questions, design, and... (part 4)

The first question is answered algebraically and by machine-precision checks; the others are conditional simulation results. The independent unit, resampling scheme, and fixed hidden-state/attack-target estimand are specified in sec. 8.22 and sec. 8.27.

How to read the visual architecture

The manuscript uses two complementary visual layers. The formal schematics adapt the generative-model and posterior-sharing perspective of Friston et al. (2024) to the categorical implementation: fig. 7 shows the private sensory report and $A/B/C/D$ substrate,

How to read the visual... (part 2)

fig. 6 shows how three local posterior messages become server inputs, and fig. 8 shows the hidden-state, agent, and active-control context. These diagrams are explanatory maps, not additional empirical observations.

How to read the visual... (part 3)

The data-bearing figures then report the executed recovery checks, conditional contamination sweep, and objective/descent diagnostics; their captions identify the relevant uncertainty and resampling unit.

How to read the visual... (part 4)

The graphical abstract in fig. 4 compresses the same logic into a recovery anchor, a federation pathway, and three explicitly non-transferable robustness axes.

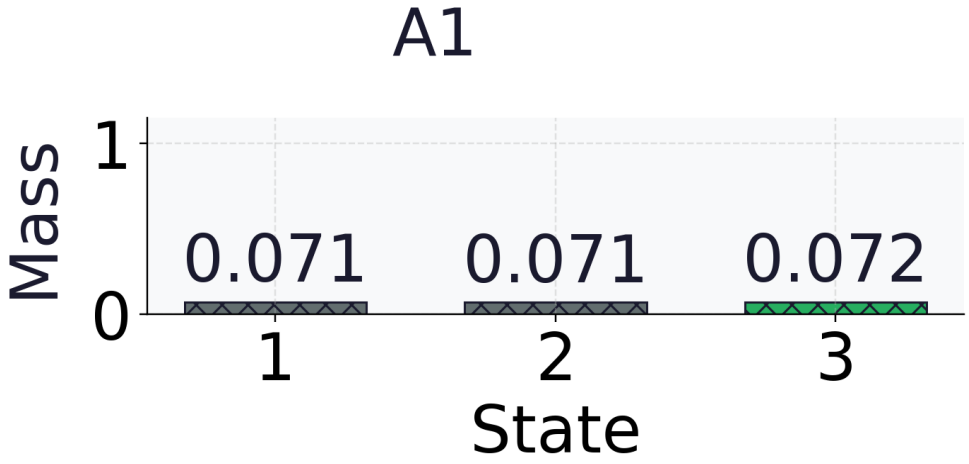
How to read the visual... (part 5)

fig. 5 illustrates one configured failure-and-repair case: a partially contaminated colony broadcasts beliefs (Panel A), the equal-weight log-linear pool is pulled toward the attack target (Panel B), and the server-side heuristic reweights the broadcasts (Panel C). It is a deterministic schematic,

How to read the visual... (part 6)

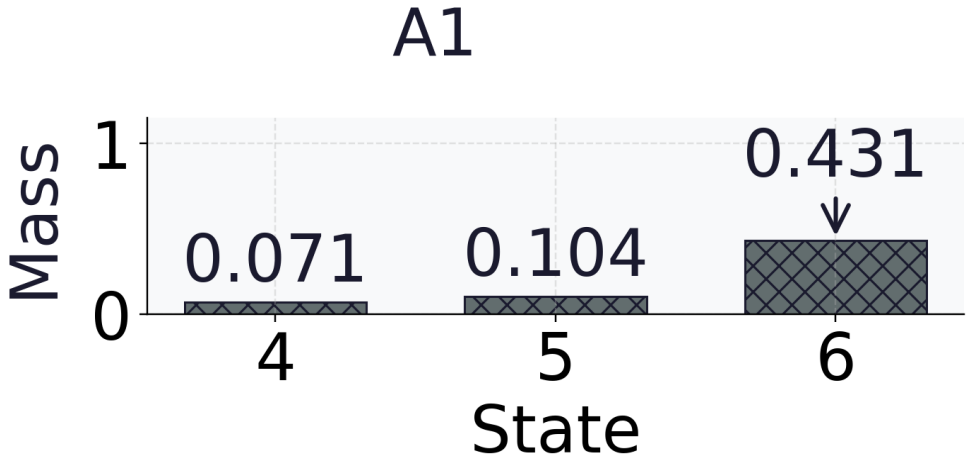
not a universal robustness claim; the three axes and their distinct guarantees are defined in sec. 6 and sec. 6.3.

How to read the visual... (part 7)



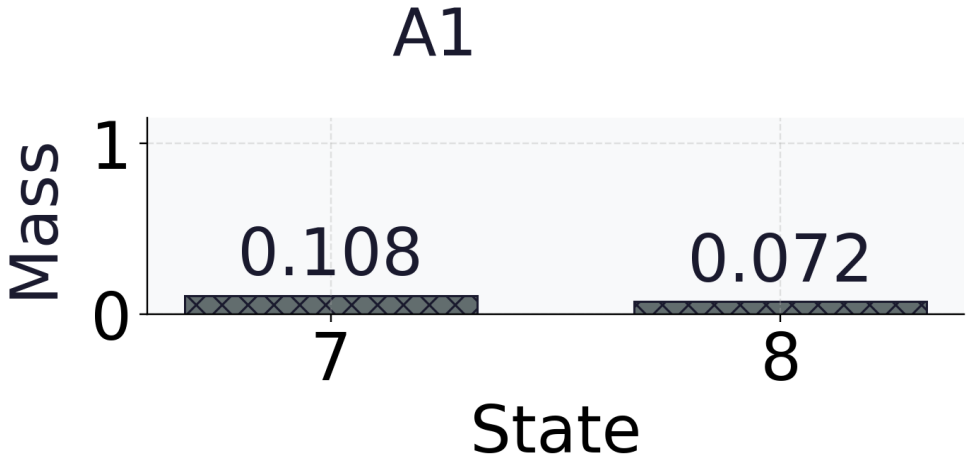
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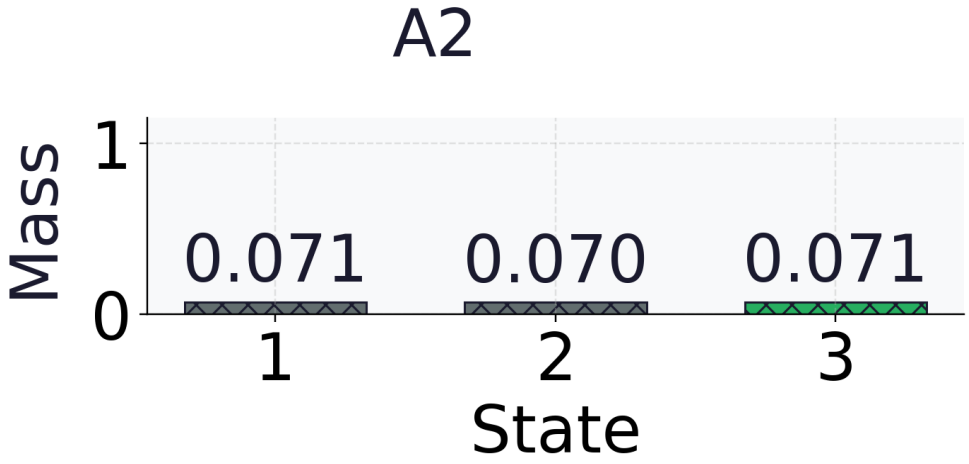


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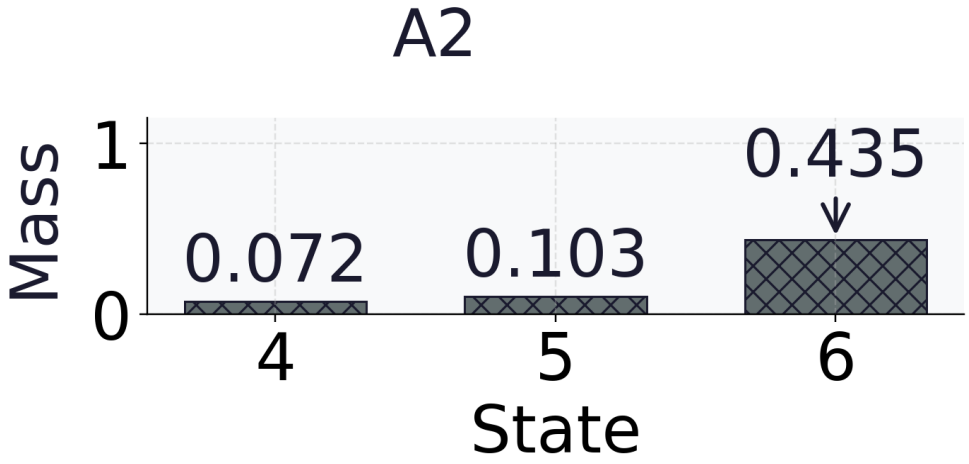


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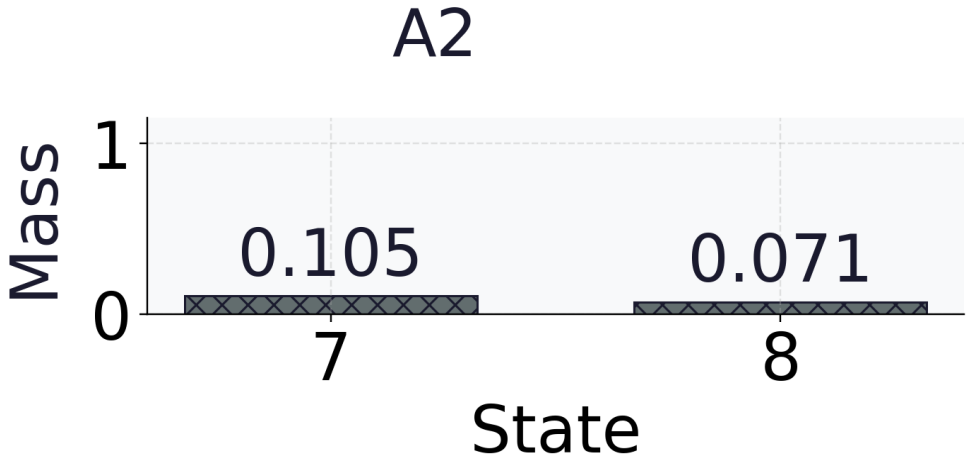
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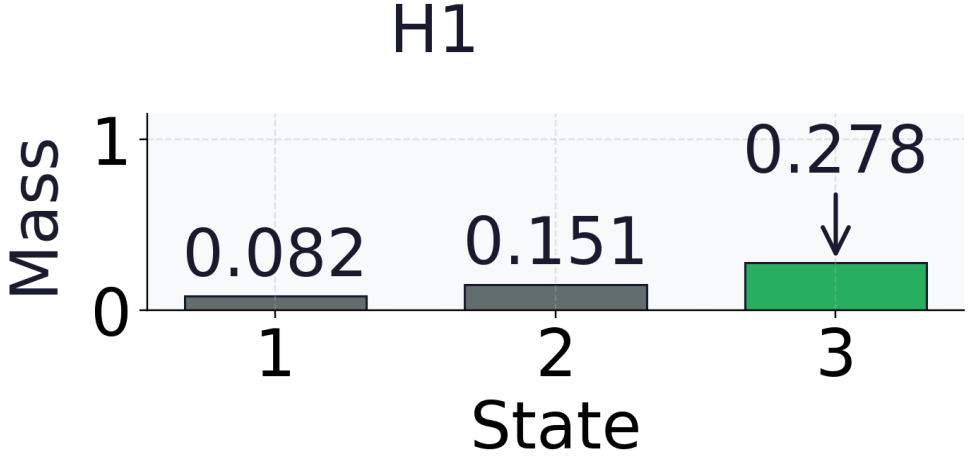


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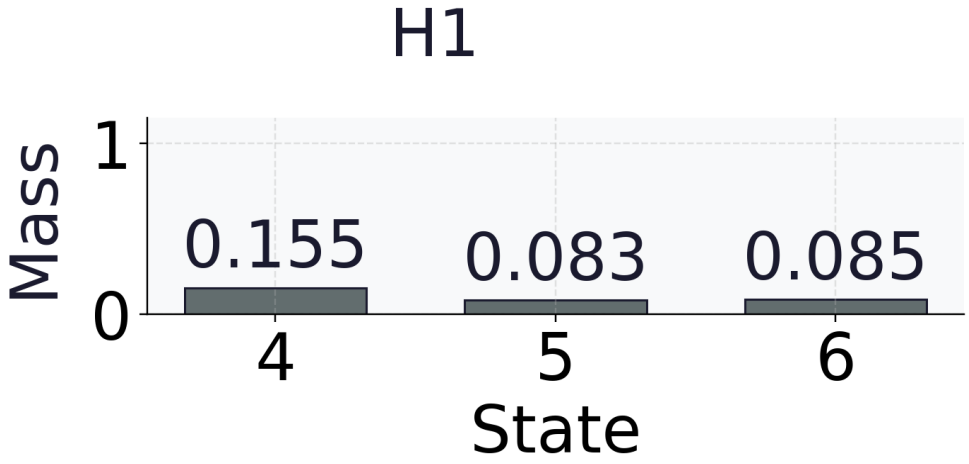
How to read the visual... (part 12)



How to read the visual... (part 13)

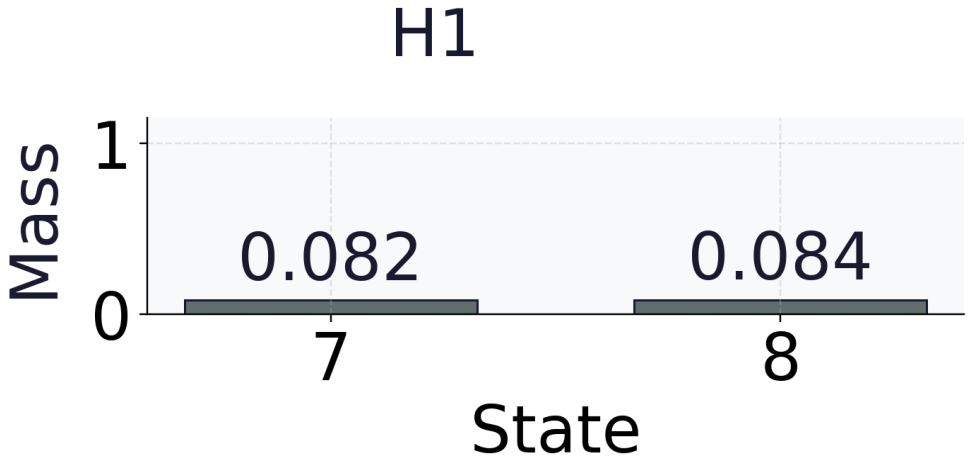


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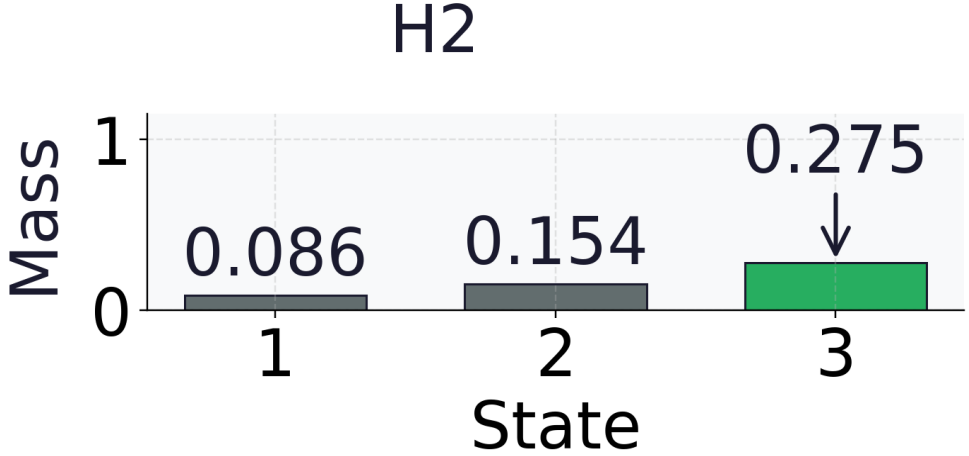


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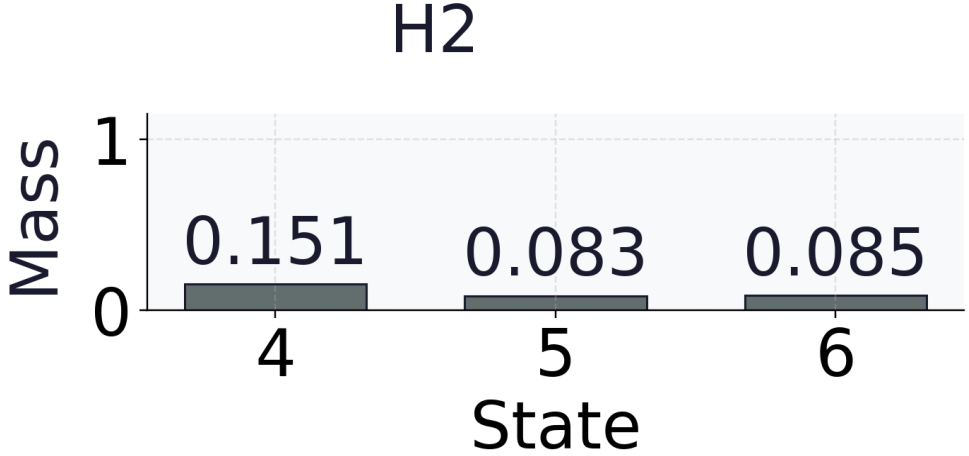
How to read the visual... (part 15)



How to read the visual... (part 16)

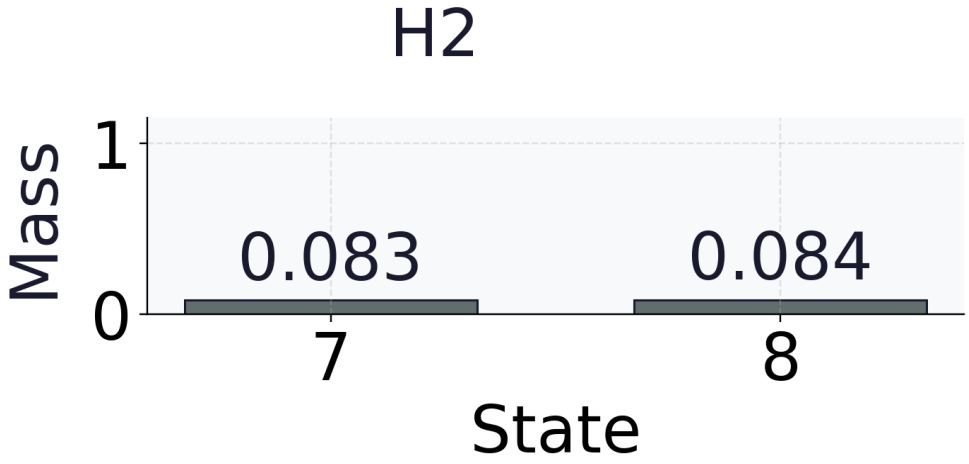


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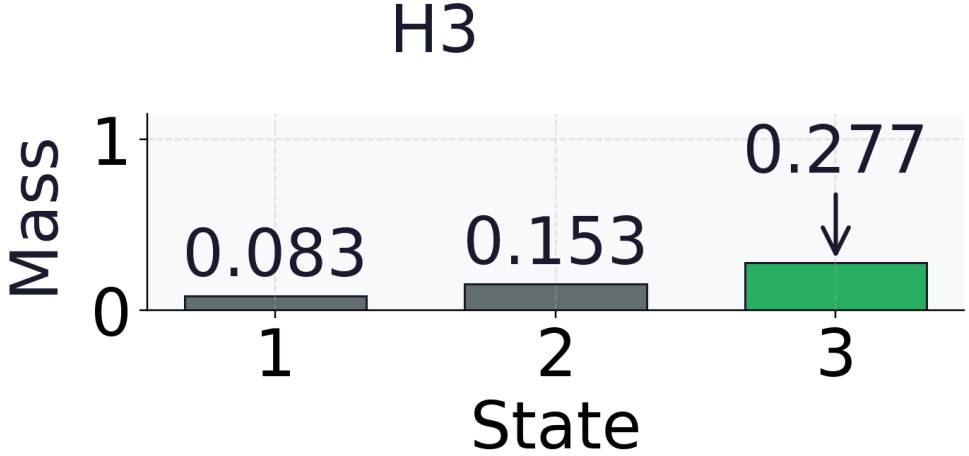


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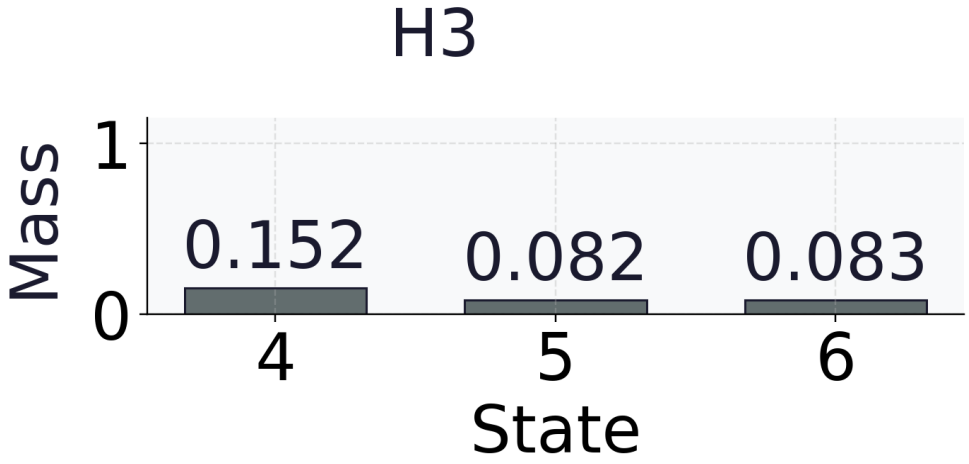
How to read the visual... (part 18)



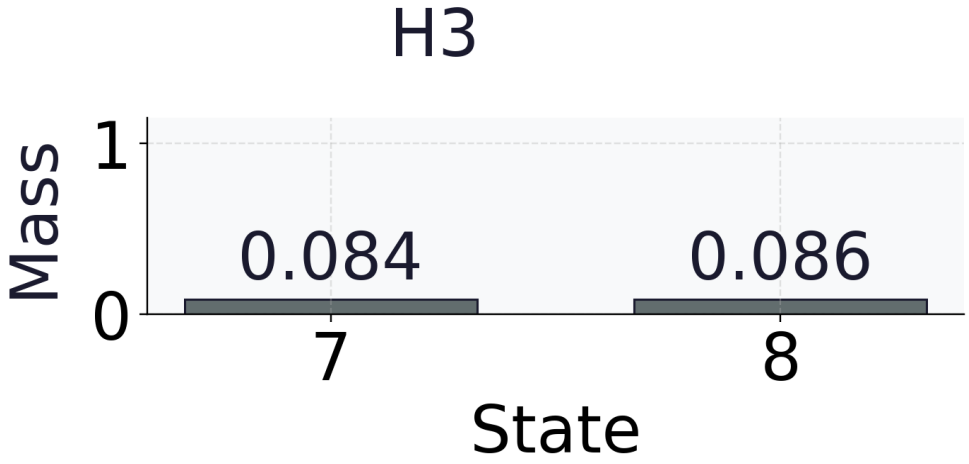
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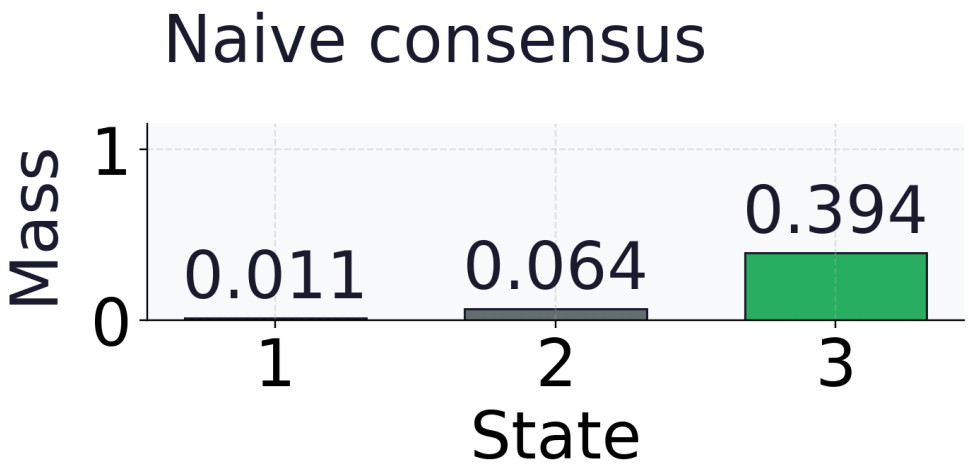
How to read the visual... (part 20)



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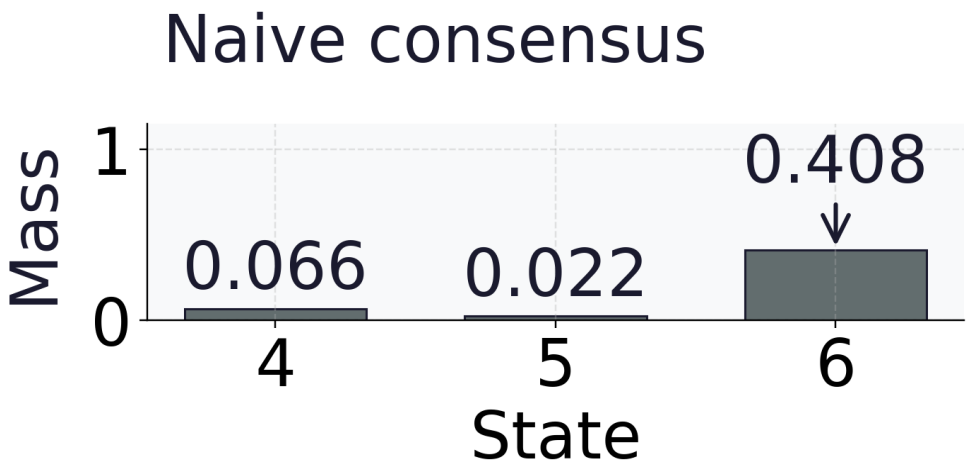


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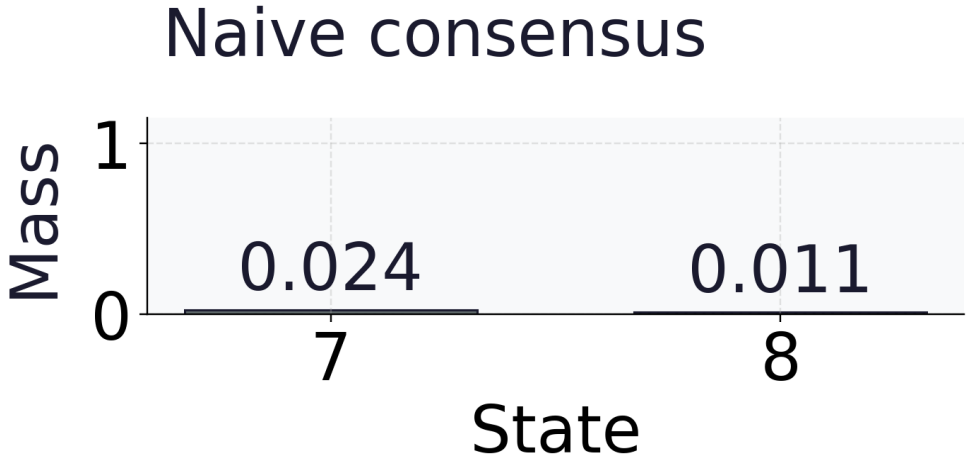
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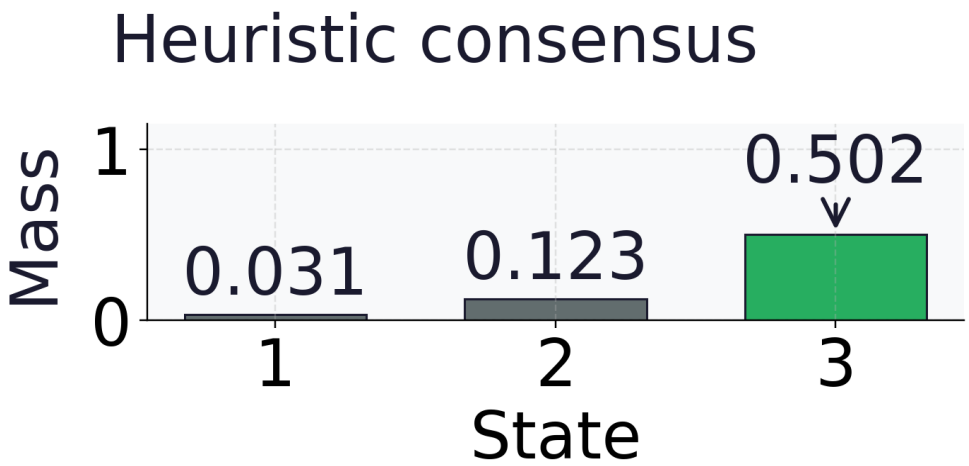
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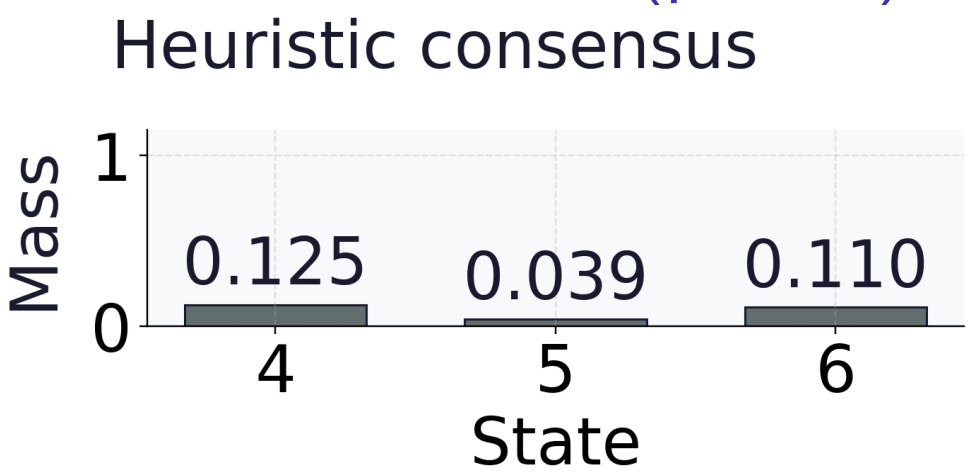
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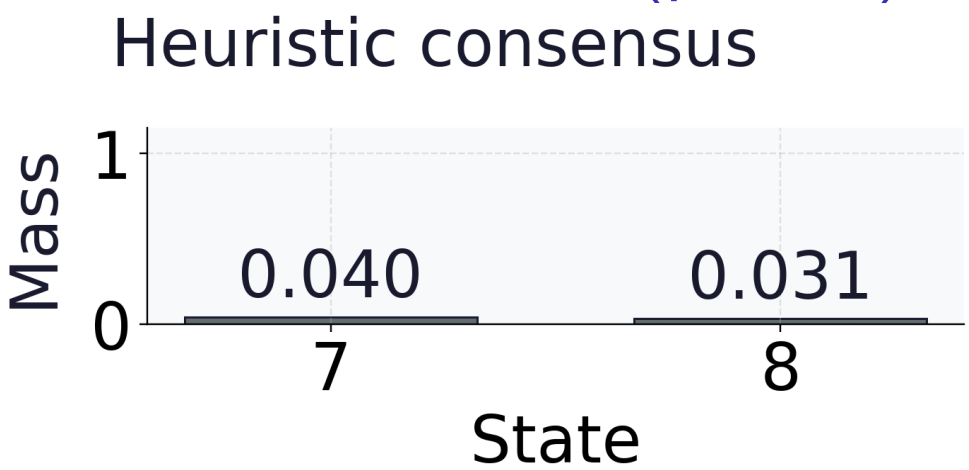
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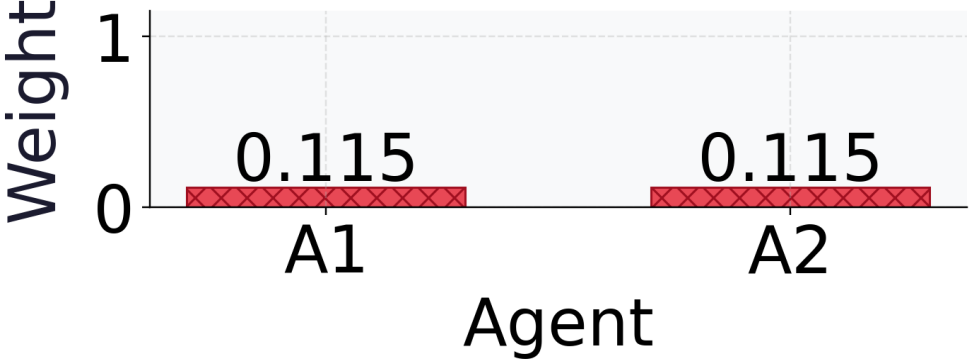
How to read the visual... (part 27)



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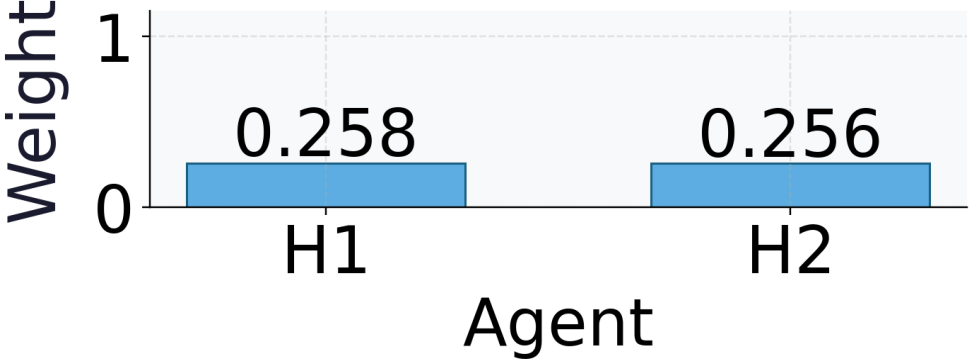
Effective influence weights



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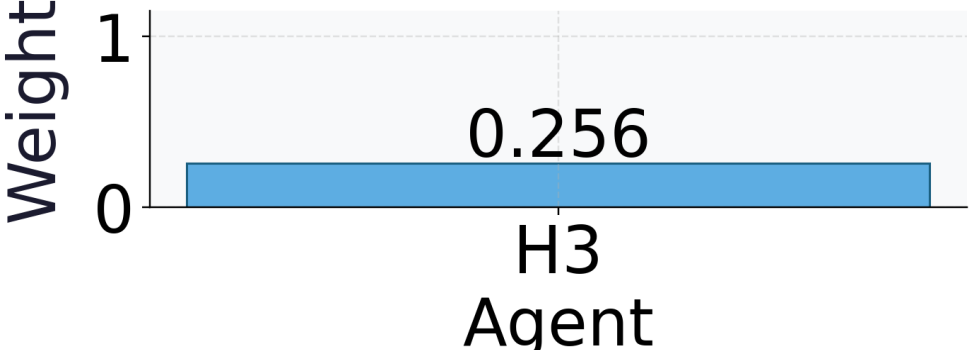
Effective influence weights



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How to read the visual... (part 30)

Effective influence weights



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How to read the visual... (part 31)

Configured colony: 5 agents; 2 adversarial, 3 honest. Contamination: 40%. True state: 3.

How to read the visual... (part 32)

Naive equal-weight log pool:
true-state mass 39%; argmax state 6.

How to read the visual... (part 33)

Heuristic robust aggregation:
true-state mass 50%; argmax state 3.
Robustness 1.5.

How to read the visual… (part 34)

A / crosshatching: adversarial; H / plain: honest. All probability panels share a 0–1 scale. The true state is green; other states are gray.

How to read the visual… (part 35)

One deterministic configured colony, not a replicated benchmark. True-state mass percentages are not deployment accuracy estimates. No uncertainty interval applies.